

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 40

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Harvest rate below the maximum sustainable yield; only economically valuable species (mahogany) selectively logged, not clear-felled • Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing; large seed trees and small trees left to grow • Buffer zones with no harvesting or machinery near rivers / wetlands • Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees • Each felled tree tagged and tracked to the retailer for FSC accreditation; rangers control illegal logging; tree nurseries replant over-exploited and ecologically important species	5
01.2	• The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it • Employment as rangers, forestry and tourism workers; ecotourists use local restaurants, produce and souvenirs; offcuts given to local carvers • Benefits give the community an incentive to protect the forest; carbon sequestration payments from companies fund management	3
01.3	• Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept; timber resource used over a longer period • Disadvantage: more labour-intensive and expensive; more tracks needed per tree	2
Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.		

Question	Marking guidance	Marks
02.1	• Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish • Species occupy different niches / trophic levels so they do not compete and all food sources are used • Bottom feeders re-suspend nutrients boosting plankton / plant growth • Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species	4
02.2	• IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) • Wastes of one species become nutrients for another so less pollution and extra harvests • Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) • Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar	4
02.3	• Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages • Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
03.1	• Sigmoidal (S-shaped) growth curve • Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate • Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate • Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate	4
03.2	• Density-dependent: effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation • Density-independent: chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter	2
03.3	• Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density • At this density deer over-graze, preventing tree regeneration and damaging habitats / other species; culling keeps numbers below the damaging level (as in Scotland)	2
03.4	• Rising prey numbers give predators more food so predator numbers rise; high predation then reduces the prey population • Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	

Question	Marking guidance	Marks
04.1	• Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors; many rare species with small ranges • High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) • Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches; intense competition for light drives specialisation • Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships; plants flower and fruit at different times	4
04.2	• Adapted to stable conditions so they have narrow ranges of tolerance; a small change can move conditions outside the range • Many species are rare with small ranges / small populations	2
04.3	• No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) • Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril)	2
04.4	• Over half the Amazon's rainfall comes from transpiration within the forest; clearance reduces downwind rainfall • Rapid decay recycles nutrients so soils are shallow / poor; canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	
05.1	• Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade • Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae • Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish)	6
05.2	• Reduces light penetration so submerged plants cannot photosynthesise • Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed	2
05.3	• Rare species often have very specialised habitat requirements / narrow ranges of tolerance • A small change moves conditions outside their range while generalists survive; small populations cannot recover	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
06.1	<i>Any 2 from:</i> • Potato bean (N America): high protein content • Morama bean (southern Africa): protein content similar to soya, drought resistant • Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils • Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion • American bison: graze grassland too poor for cattle • Common eland: browse and graze so can be kept with cattle without severe competition	2
06.2	• Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil • So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted	2
06.3	• They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
07.1	• Mars has low mass so weak gravity and could not retain a dense atmosphere • Very low atmospheric pressure means water would boil / evaporate away even at low temperatures • Greater distance from the Sun gives low insolation and a mean temperature of -60 °C so water is frozen	3
07.2	• Mass great enough for gravity to retain an atmosphere containing the elements needed (C, H, O, N as CO₂, methane, ammonia, water vapour) and to maintain pressure so liquid water does not boil • Distance from the Sun (the 'Goldilocks zone') gives insolation producing temperatures at which water is liquid (0-35 °C over most of the surface) and proteins are not denatured • Too close (Venus) - too hot; too far (Mars) - too cold	3
07.3	• The molten core produces a magnetosphere that deflects the solar wind / charged particles, preventing biologically damaging radiation reaching the surface • The 24-hour rotation limits heating by day and cooling by night, reducing temperature extremes (Venus' slow rotation gives extreme conditions)	2
07.4	• The tilted axis causes seasonal variation in insolation as the Earth orbits the Sun • Producing seasons (temperature, day length) to which species' life cycles (breeding, migration, dormancy) are adapted; the Moon's gravity keeps the tilt stable and creates tides	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
08.1	• There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change • Random sampling might leave gaps / miss the pattern; systematic sampling avoids bias while covering the gradient	2

Question	Marking guidance	Marks
08.2	• Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data; wider intervals (eg 4 m) may miss peaks and troughs • Take increasing numbers of samples at one site and calculate the running mean; the number needed is where the mean stops fluctuating / stabilises • Calculate standard deviation to assess scatter: more variable data need more samples • Test the method (extraction technique, timing) for reliability before the main study	4
08.3	• Remove surface vegetation; pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates • Collect worms that come to the surface for a standardised time; count / identify / weigh • Limitations: solution may not percolate impermeable soils or reach deep worms; worms may move sideways or downwards; some may die or not respond; slow-moving worms missed	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
09.1	• $600 + (90 + 150) - (220 + 80) = 540$ • Stock fell by 60 so the catch exceeds the MSY / unsustainable	2
09.2	• $A + G - M = 90 + 150 - 80 = 160$	1
09.3	• Catch is landed and weighed at ports so it is recorded • Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed • Representative sampling of a large enough proportion of the area is impractical / expensive; survival of eggs and larvae varies greatly between years	3
09.4	• CPUE: if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained; a better measure than catch size which technology can prop up • Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
10.2	• Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
11.1	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why monitor: species present, population size, density, distribution, survival, age structure, breeding rates, movements; baseline and evaluation; Red List categorisation and quotas rely on it • Standard techniques and their limitations: quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR; sampling design (random / systematic, size, number, timing, standardisation) • New technologies: satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers • How they improve validity: representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity • Examples: penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked • Limitations: cost, battery / tag weight, data handling, ethics, still need ground truthing • Judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7.	25